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Speech Emotion Based Music Recommendation

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***Abstract*—Music has always been a very popular form of entertainment and many pioneers in the tech industry have quickly realised this. With the growth of music streaming platforms, users have a multitude of options to choose from. This calls for a system that makes music organization and search management easier. This work uses Siamese network, a meta learning model, to recognize emotions in speech, based on which music recommendations are made. We have used RAVDESS dataset for emotion recognition and music mood classification dataset. The user emotions are mapped to appropriate moods of songs and used for recommendations.**

***Index Terms*—speech emotion recognition, siamese network, meta learning**

I. Introduction

Speech is a very important part of human communication and is used to convey both linguistic and paralinguistic information. Classical speech recognition relies on retrieval of only linguistic information and fails to identify the paralinguistic information such as gender, emotion and state of mind of the speaker. The original form of emotion recognition was based on recognizing emotions from facial expressions. Speech Emotion Recognition (SER) is an area of research that has gained a lot of popularity in recent times. Speech Emotion Recognition (SER) is the process of recognizing emotions from speech signals. SER has become an important domain of research over the years due to the popularity of voice-based search in devices such as Google Home and Amazon Echo. To recognize the emotions in speech, the audio signals are first converted into spectrograms. Spectrograms are the visual representation of frequencies of the signal with respect to time, where the audio intensities are expressed in terms of colors i.e., a bright color signifies a heavily concentrated sound, whereas a dark color represents a dead sound.

The main challenge of SER is that the models usually have a low accuracy, bordering at 60%, when trained using conventional deep learning models such as Convolutional Neural Network (CNN) and Recurrent Neural Network (RNN). The reason for this is that SER relies on a huge sample size to learn effectively. Besides, speech emotion varies with gender, race and region. Since having a huge dataset is neither cost effective nor practical, this paper attempts to train a model to recognize emotions in speech using meta learning. Meta learning techniques are effective in training a model with a limited amount of data.

Meta learning is a class of machine learning algorithms that learn how to learn across a number of different prediction experiments. It is a class of algorithms that observes how different ML algorithms learn during training and then learn from this "experience". In this class of learning the model relies on a lesser number of training data and is less prone to overfitting. The different types of meta learning models are: Metric based, Model based and Optimization.

This work uses a Siamese Network which is a metric based learning algorithm. Siamese network is a one shot classification model that aims to make classifications by training on a single example per class. For this experiment, however, each class has more than one example per class. Siamese network is a set of 2 similar networks that have the same architecture, parameters and weights. The base network is formed by a convolutional neural network, to which the input images are based in batches. The data is first split into positive and negative pairs. Positive pairs consist of similar images whereas the negative pairs are a pair of dissimilar images. The networks output feature encoding of the 2 input images at the final fully connected layer of the base network.

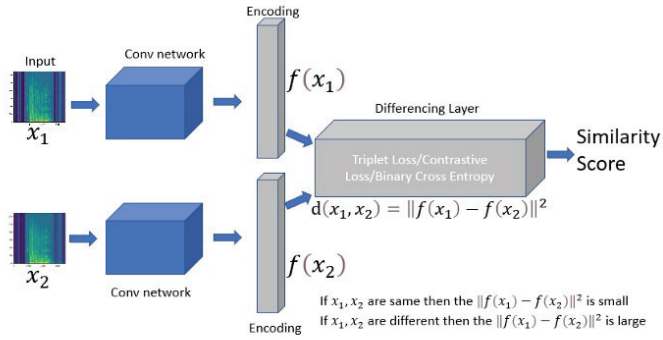

Fig. 1. Siamese Network

The difference between the outputs can be calculated by using a loss function such as contrastive loss, binary cross entropy or triplet loss function. For the purpose of this experiment, contrastive loss function was used.

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By contrasting the inputs, the contrastive loss function distinguishes between similar and dissimilar inputs. The goal of the contrastive loss function is to reduce the distance between positive pairs while increasing the distance between negative pairs, promoting intra-class compactness. The distance between a positive instance and another instance of the same class is calculated and compared to the distance between negative examples in contrastive loss. This is done by calculating the vectors' cosine distances and interpreting the resulting distances as prediction probabilities from a standard classification network. The main notion is that you may utilise cross entropy loss to treat the distance of the positive example and the distances of the negative instances as output probabilities.

The recommendation module is responsible for recommending music to the users based on their personal preferences and current emotion. This module uses a dataset that has a list of songs along with the song features such as energy, danceability, valence, arousal etc, based on which the songs have been mapped to appropriate moods. These moods are mapped to the appropriate human emotions and the corresponding songs are recommended. A feedback system is used to keep track of the user's preference to aid future recommendations.

The rest of this paper is organized as follows: Section II elaborates on the research done in the field of SER and emotion based recommendation. Section III explains the Proposed Methodology and Section IV discusses the experimental setup in detail. This is followed by Conclusion and Future Work in Section V.

II. Related Work

A. Speech Emotion Recognition

The existing work for speech emotion recognition includes models using Support Vector Machine (SVM), deep neural network architectures such as 1-dimensional Convolutional Neural Network (CNN) and 3-dimensional Convolutional Recurrent Neural Network with Attention Model (3D-ACRNN). Two-dimensional CNN has proved to be efficient in working with audio spectrograms. Modulation spectral features and MFCC were used to obtain the emotional features. For each frame, the Fourier Transform and energy spectrum were calculated and translated into mel scale. These extracted features were given as input to a seven layer convolutional neural network followed by a Batch Normalization and Max Pooling layer. The model could achieve an accuracy of 83.61% on SAVEE dataset.

In another approach, a 3 dimensional attention-based CRNN (Convolutional Recurrent Neural Network) was used for SER. In this work, the deltas and delta-deltas of log-Mels of the input speech signals were calculated and were given as input to CNN architecture. These 3D CNN features were fed into LSTM, followed by an attention layer. The attention layer produces utterance level features as output from the input high level features. These features are finally fed into a fully connected layer. The model achieved an accuracy of 62.825% on IEMOCAP dataset.

B. Meta-Learning in Speech Emotion Recognition

Zheng Lian et al. presented a model for SER using Siamese Network and contrastive loss to learn discriminative audio representations. The loss function used was a combination of contrastive loss and cross entropy loss. The architecture consists of 2 similar networks with the same parameters and weights. The speech signals are split into s overlapping frames of d dimensions and this sxd matrix is fed into the CNN base network. The loss is calculated by combining the contrastive and cross entropy loss using weighting coefficient. The experiment was conducted for Feature Level Comparison, Multi task learning and to compare the impact of contrastive loss. The proposed method obtained an accuracy of 63.21% in unweighted accuracy and 62.19% in weighted accuracy on IEMOCAP dataset.

In another work, Suransh Chopra et al. Model Agnostic Meta-Learning (MAML) was used for SER in low resource languages. The effect of meta learning was studied on five different combinations of source datasets: TESS, EMODB, RAVDESS, SAVEE and IEMOCAP. The base model was a 2 layer LSTM with 2 dense layers at the end. The experiment is done using two optimization methods - meta updates and task specific updates. The MAML based approach required rigorous hyper-parameter tuning and was prone to errors during training.

C. Emotion Based Music Recommendation

Shlok Gilda et al. presented a music mood recommendation system that recommends music as per the emotions detected from the facial emotions of the user. The model was divided into 3 modules: Emotion Module, Music Classification Module and Recommendation Module. The Emotion module was trained using the Kaggle FER2013 dataset. They used a 2D convolutional neural network to detect facial emotions with an accuracy of 90.23%. The Music classification module classified songs into 4 different moods based on spectral and rhythmic features. The MLP architecture achieved an accuracy of 97.69%. The Recommendation module mapped the user emotions to appropriate song moods and made recommendations based on user preference. The module was developed using Russell 2D Valence-Arousal Model and Geneva Emotion Wheel.

Since recommending songs based on emotions derived from speech is relatively less popular, owing to the need for a large dataset, this work aims to build speech emotion based music recommendation system using meta learning, that will rely on minimal data to effectively recognize the emotions from speech and make recommendations accordingly. The recommendation system built also takes into account the user's feedback to improve the quality of recommendations in the future. In this work, Siamese Network has been trained using 4 shots in each emotion class to obtain the best accuracy.

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D. Novelties and Contributions

Following are the key contributions of our work:
1. In the proposed work we have developed a model with 6 CNN and 3 dense layers that was shown to yield the maximum accuracy in recognising the emotions from speech inputs.
2. Siamese network, a one shot learning technique, along with contrastive loss function is used to reduce the dependence of the model on huge amounts of training data. In this work the model has been trained using 4 training examples per class.
3. The results from the recommendation module is evaluated using 10-fold cross validation.
4. In addition, we have also tried to find out the optimum values for parameters such as hanning window size, kernel size, number of layers in the model and stride in order to reduce the complexity of the model and obtain maximum accuracy.

III. Proposed Methodology

The underlying methodology used to implement an SER based music recommendation system is explained in this section. Since SER relies on large amounts of data to give an acceptable accuracy, we have employed Siamese network, a meta learning model to learn and identify the emotions in the speech input. Siamese network is ideally a one-shot learning problem, where the model is trained using a single example from each resultant class. However, for this model, Siamese network is being implemented as a few shot learning model. The Siamese model architecture used here consists of a series of two-dimensional convolutional and max pooling layers followed by three dense layers that will result in a 64D feature vector. The emotions identified using this model are passed to the recommendation module, which is implemented using a three layer MLP architecture with 10 fold cross validation to obtain optimum results.

A. Classification using Siamese Network

Siamese Networks are popularly used for one shot learning. One-shot learning refers to classification tasks in which a large number of predictions must be made based on only one (or a few) instances of each class.
Given a support set S and a test image x, the final predicted class is:

$$\hat{c}_S(x) = c(argmax_{x_i \varepsilon S} P(x, x_i)) \tag{1}$$

where c(x) is the class label of an image x and c(.) is the predicted label.
For the purpose of this experiment, we are training the model on more than one instance of each class.
Each "branch" in the network computes the feature vector of the input image and the similarity of the features is then computed. For images belonging to the same class, the target output is 1, else 0. To find the similarity between these vectors we use contrastive loss. The distance between 2 points is measured as follows:

$$D_w(X_1, X_2) = ||G_w(X_1) - G_w(X_1)|| \tag{2}$$

Here, X_1 and X_2 are the first and second inputs given to the Siamese network. $G_w(X_1)$ and $G_w(X_2)$ are points that result from mapping X_1 and X_2. The contrastive function must return a small value, if X_1 and X_2 belong to the same class. Else the value is high. The loss function is defined as:

$$L(W) = \sum_{i=1}^{N} L(W, (Y, X_1, X_2)^i)) \tag{3}$$

$$L(W, (Y, X_1, X_2)^i) = \begin{cases} L_G(D_w(X_1, X_2)^i) Y = 1 \\ L_I(D_w(X_1, X_2)^i) Y = 0 \end{cases} \tag{4}$$

where $(Y, X_1, X_2)^i)$ is the i-th sample, L_I is the loss function for negative pairs and L_G is the loss function for positive pairs.

B. Network Architecture

Siamese network uses two twin CNN networks having similar weights and parameters. Both the networks learn an efficient embedding to get the relationship between the input images. The base network consisted of 6 convolutional layers and 3 dense layers and a final normalised layer that converts the output of the final dense layer into normalised data. These networks are found to be robust to class imbalance problems and can train themselves using minimal data per class. All the layers use ReLu (Rectified Linear Unit) activation function. The max pooling layers have a pool size of 3x3 and stride set to 1x1. The dense layers have 256, 128 and 64 units respectively.
The MLP architecture used for the recommendation module had an input layer with 8 units and activation function ReLu, followed by 2 dense layers having 6 and 4 units respectively and softmax activation function. The model used categorical crossentropy and Adam optimizer.
These choices have been made after various experiments to improve the model accuracy.

C. Dataset Description

The model uses RAVDESS (Ryerson Audio Visual Database of Emotional Speech and Song) dataset for Speech Emotion Recognition. The dataset is trimmed to 4 actors (2 male and 2 female) displaying 8 emotions – happy, sad, anger, disgust, surprise, calm, fear and neutral. There are two levels of intensity for each expression: normal and strong. These data are reorganized into 8 different directories, each directory representing a class of emotion, making it easier for the Siamese network to learn.
For the recommendation module, we use a 'data_moods'

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dataset that has a list of over 600 songs obtained using the Spotify API, along with features such as popularity, length, danceability, accousticness, valence, energy, loudness etc. This data is trained using a Valence Arousal model and each song is mapped to one of the 4 moods - Happy, Serene, Energetic, Calm.

D. Data Preprocessing

The input data is in the form of speech signals. Signals are ideally represented using waveforms. Signals are ideally represented using waveforms, as shown in Fig 2.

In order to work with these signals, they need to be first

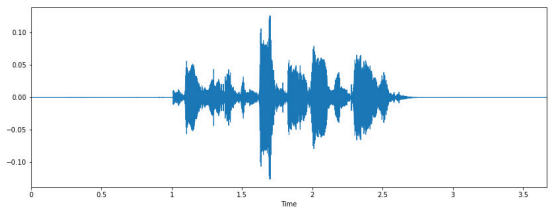

Fig. 2. Representation of an audio waveform

converted into visual representations of the same, referred to as spectrograms. To obtain these spectrograms, the audio data is first split into windows. The window size is set to 512, as opposed to a default of 2048 as this is found to be ideal for speech signals. We use Hanning windows for this work as it is proven to provide good frequency resolution and reduces spectral leakage. We performed Short Term Fourier Transform (STFT) on each window and took the absolute values and averaged them together. This results in each window being represented using a vertical line that represents magnitude vs frequency. These values are then converted into decibels, a perpetually relevant representation of frequencies that is logarithmic in nature. These windows are laid out and displayed as an image. Fig 3 shows the spectrogram obtained for the audio waveform in Fig 2.

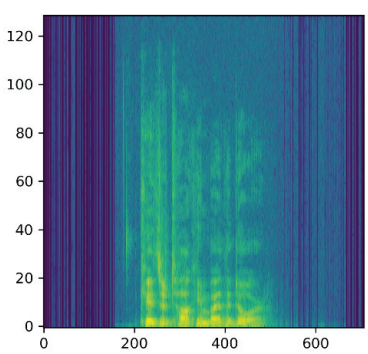

Fig. 3. Spectrogram of an audio sample

The data_moods dataset is preprocessed to include an additional column 'Watch_Time' that keeps track of the user interaction or feedback. If the user considers the recommendation being made and chooses to listen to the song, the value of Watch_Time is upgraded, thereby increasing the song's probability to be recommended once again. The model is trained based on features that have a high correlation with the target column.

IV. Experimental Setup And Results

The proposed work consists of two modules – Emotion Recognition Module and Recommendation Module. The flow of control in the proposed system may be depicted as follows:

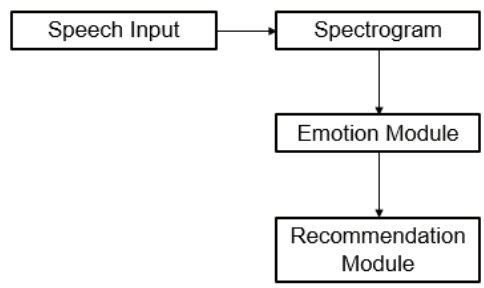

Fig. 4. Flow of control

A. Emotion Recognition Module

Emotion Module is responsible for extracting the emotion information from the user's speech input signals. A Siamese network is used for this purpose. The audio data is preprocessed to extract the spectrogram of the speech signals. These spectrogram plots are saved as images in the system and are used for training the network.

The speech signals are given as input to the Emotion Recognition Module, which uses Siamese network with contrastive loss to recongize the emotions in the speech. The signals are first converted into a spectrogram by applying hanning window on the audio data which are tested with the existing data and categorized into any one of the 8 available emotion classes, based on the loss value. The model uses four-shot learning technique to perform the classification.

The model was tested for various number of iterations and variations in parameters, such as varying window size, batch size and image size of the spectrogram and yielded an accuracy of 71.43% for a base network with 6 CNN layers and 3 dense layers where 85% of the samples were used for training and 15% for testing. Setting a small batch size such as 8 allows the entire dataset (of 32) samples to be processed in a nominal number of batches and proves to be more effective than larger numbers. Concentrating the features into a 64 dimensional vector using 3 dense layers was also shown to improve the accuracy.

Table I shows a comparison of results of the proposed method

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with the existing literature survey for emotion recognition.

TABLE I. COMPARISON OF RESULTS WITH LITERATURE SURVEY

Author [ref.]	Architecture used	Dataset Used	Accuracy Obtained
Zheng Lian [4]	Siamese Network + Contrastive Loss	IEMOCAP	62.19%
Tae-Yeun Kim [2]	SVM	Korean-style emotion speech database	86.98%
Alif Bin [13]	CNN	RAVDESS, IEMOCAP	43.4%, 83.61%
Shlok Gilda [1]	CNN	Facial Expression Recognition Challenge dataset	90.23%

TABLE II. ACCURACY (%) FOR VARIOUS PARAMETERS

Architecture	Iterations	Batch Size	Accuracy
6 CNN + 2 dense	20	8	70.0%
6 CNN + 2 dense	30	8	50.0%
6 CNN + 2 dense	20	16	57.0%
6 CNN + 3 dense	20	8	71.43%
6 CNN + 3 dense	20	16	56.67%
8 CNN + 2 dense	20	8	25%

B. Recommendation Module

This model used a dataset data_moods.csv which contained a list of songs along with their Spotify ids and features obtained using the Spotify API. These features were used to classify the songs into 4 different moods - Happy, Serene, Calm and Energetic, using Valence Arousal model. These moods are mapped to the human emotions from the SER module as follows:

TABLE III. MAPPING USER EMOTIONS TO MUSIC MOOD

Music mood	User Emotion
Happy	Happy
Serene	Sad
Calm	Energetic, Surprise, Fearful, Angry, Disgust
Energetic	Neutral

The model consists of a 2 layer MLP architecture that used 10 fold cross validation where 80% of the samples were utilised for training and 20% for testing. The model yielded an accuracy of 97.82%. The system records the user feedback/interaction each time and utilises this to make future recommendations.

The table below shows the various experiments done conducted to obtain the optimum accuracy.

TABLE IV. ACCURACY (%) FOR VARIOUS PARAMETERS

Architecture	Iterations	Batch Size	Accuracy
1 hidden layer	300	200	74.0%
2 hidden layers	300	200	97.82%
3 hidden layers	300	200	76.80%

V. Conclusion and Future Work

With the rise in popularity of devices such as Google Home and Amazon Echo, voice-based searching and speech recognition will go on to become the key points of research and profits in the world of technology. It is therefore, necessary that the current systems begin to incorporate these into their applications to survive in the long run. Music streaming platforms are a huge source of profit in the tech industry. Funneling the search, stream and recommendation of music based on user preferences and more importantly, their current state of matter can build a strong and loyal customer base and keep the platform ahead of their competitors in the future. This work therefore, explores the same area using techniques of Deep learning and Meta learning to be able to work with minimum data with maximum accuracy. In this work, we have found the ideal value for various hyper-parameters in order to achieve high accuracy. The Emotion module employed 4-shot learning method on 32 samples displaying 8 basic emotions to achieve an overall accuracy of 71.43%. The usage of meta learning has reduced the dependency of the SER model on large amounts of data, as compared to the existing work and has given a significantly higher accuracy. The Recommendation module used 10-fold cross validation and relied on a dataset of over 600 songs mapped to appropriate moods, to make recommendations based on the user's emotion extracted from the speech input. The model was trained using a batch size of 200 and ReLu activation function and gave an accuracy of 97.82%.

The model can further be improvised by the use of meta learning techniques for recommendation module. This will reduce the dependence of the recommendation module on large amounts of dataset and will also address the cold start problem. In future, music recommendation can be made using bimodal or multi modal emotion recognition, based on emotions recognised from body gestures and facial expressions. This will significantly reduce the variations in results due to difference in gender, race, region etc.

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